Quantile Regression

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Let $(X_i', Y_i)'$ for $i = 1, \ldots, N$ be a random sample of covariate, X_i, and outcome, Y_i, draws from the joint distribution $F_{X,Y}$. As a concrete example, Y might equal log-earnings and X a vector including years of completed schooling, age and other proxies for human capital. Following Chamberlain (1994), we will assume the X is discretely-valued with support $\mathbb{X} = \{x_1, \ldots, x_M\}$; x_m might, for example, correspond to a particular configuration of schooling and age.

Conditional quantiles

Let $F_{Y|X}(y|\, X = x_m) = \Pr(Y \leq y|\, X = x_m)$ be the conditional distribution of log-earnings among the subpopulation of individuals with $X = x_m$. The corresponding conditional quantile is

$$Q_{Y|X}(\tau|\, X = x_m) = F_{Y|X}^{-1}(\tau|\, X = x_m). \tag{1}$$

As a function of $x \in \mathbb{X}$, equation (1) is called the τth *conditional quantile function* (CQF) of Y given X. For a given τ, the CQF takes one of only M possible values: one for each point in the support of X.

Let $N_m = \sum_{i=1}^{N} \mathbf{1}(X_i = x_m)$ equal the number of sampled units belonging to the $X = x_m$ sub-population. Let $W_{(1)}, W_{(2)}, \ldots, W_{(N_m)}$ denote the order statistics of Y *within this sub-sample*. We can estimate $\pi_m(\tau) \overset{def}{\equiv} Q_{Y|X}(\tau|\, X = x_m)$, analogously to how we proceeded in the unconditional case, by

$$\pi_m(\tau) = \frac{W_{(j)} + W_{(j+1)}}{2}$$

with j such that $\frac{j}{N_m+1} \leq \tau < \frac{j+1}{N_m+1}$.

Let $\hat{p}_m = N_m/N$ equal the fraction of sampled units in subpopulation m. We will assume that as $N \to \infty$

$$\hat{p}_m \to p_m > 0$$

for all $m = 1, \ldots, M$. This implies that $N_m \to \infty$ at the same rate for all M sub-populations in our asymptotic thought experiment. Under this assumption we can set $l = z^{1-\alpha/2}\sqrt{N_m \tau (1-\tau)}$, $j = \lfloor N_m \tau - l \rfloor$ and $k = \lfloor N_m \tau + l \rfloor$ such that the interval

$$\left[W_{(j)}, W_{(k)}\right]$$

covers $\pi_m(\tau)$ with an ex ante probability of (approximately) $1-\alpha$ (remember that $W_{(j)}$ is the jth order statistics of the Y's in the $X = x_m$ subsample).

It is also true that

$$\sqrt{N_m}\left(\hat{\pi}_m(\tau) - \pi_m(\tau)\right) \xrightarrow{D} \mathcal{N}\left(0, \sigma_m^2\right)$$

for some σ_m^2. Multiplying by $\sqrt{N}$, dividing by $\sqrt{N_m}$, and recalling that $\hat{p}_m \to p_m$ yields

$$\sqrt{N}\left(\hat{\pi}_m(\tau) - \pi_m(\tau)\right) \xrightarrow{D} \mathcal{N}\left(0, \frac{\sigma_m^2}{p_m}\right).$$

We can estimate σ_m^2, adapting our method for unconditional quantiles in the obvious way, by

$$\hat{\sigma}_m^2 = \frac{N\hat{p}_m \cdot \left(W_{(k)} - W_{(j)}\right)^2}{4 \cdot \left[z^{1-\alpha/2}\right]^2}. \tag{2}$$

Finally observe that $\sqrt{N_m}\left(\hat{\pi}_m(\tau) - \pi_m(\tau)\right)$ and $\sqrt{N_{m'}}\left(\hat{\pi}_{m'}(\tau) - \pi_{m'}(\tau)\right)$, for $m \neq m'$, are conditionally independent random variables. This is because they are constructed using non-overlapping sub-samples. From this observation we get that

$$\sqrt{N}\begin{pmatrix} \hat{\pi}_1(\tau) - \pi_1(\tau) \\ \vdots \\ \hat{\pi}_M(\tau) - \pi_M(\tau) \end{pmatrix} \xrightarrow{D} \mathcal{N}\left(\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{\sigma_1^2}{p_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \frac{\sigma_M^2}{p_M} \end{pmatrix}\right). \tag{3}$$

For convenience it will be helpful to define $\pi(\tau) \overset{def}{\equiv} \left(\pi_1(\tau), \ldots, \pi_M(\tau)\right)'$ and $\Omega \overset{def}{\equiv} \operatorname{diag}\left\{\sigma_1^2/p_1, \ldots, \sigma_M^2/p_M\right\}$. We can construct an estimate of Ω using $\hat{p}_m = N_m/N$ and (2).

Quantile regression

While we can consistently estimate $\pi_m(\tau)$ for all $m = 1, \ldots, M$, when M is large we will generally require a more parsimonious summarization of the data. To this end we will assume that the conditional quantile function (CQF) is linear in a known $K \times 1$ vector of functions of X. Denote this vector by W; for example W might include a constant, years of completed

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schooling, age and age squared. We will assume that

$$\pi_m(\tau) = Q_{Y|X}(\tau | X = x_m) = w_m' \beta(\tau) \tag{4}$$

with $W = w_m$ when $X = x_m$.

Under (4) we have that

$$\hat{\pi}_m(\tau) \approx w_m' \beta(\tau),$$

where the $\approx$ is due to the fact that $\hat{\pi}_m(\tau)$ is a noisy estimate of $\pi_m(\tau)$. This suggests choosing $\hat{\beta}(\tau)$ to coincide with the solution to

$$\min_{b \in \mathbb{R}^K} \sum \frac{\hat{p}_m}{\hat{\sigma}_m^2} \left(\hat{\pi}_m(\tau) - w_m' b\right)^2. \tag{5}$$

Note that $\frac{p_m}{\sigma_m^2}$ equals the inverse of the variance of $\sqrt{N}\left(\hat{\pi}_m(\tau) - \pi_m(\tau)\right)$. This means that our criterion function (5) attempts to find a $\hat{\beta}(\tau)$ such that the fit of the linear model is good for those conditional quantiles that are precisely estimated; less emphasis is placed on goodness of fit for those conditional quantiles that are poorly estimated.

Equation (5) is similar to a weighted least squares (WLS) problem. The key difference is that the sole source of deviation between $\hat{\pi}_m(\tau)$ and $w_m' \beta(\tau)$ is due to sampling error in the former. If we knew the conditional quantiles $\pi(\tau) \overset{def}{\equiv} (\pi_1(\tau), \ldots, \pi_M(\tau))'$ exactly, then we could solve (5) to find the exact value of $\beta(\tau)$; free of any sampling error. This observation underscores the "strongness" of the linear conditional quantile regression assumption, (4) above. Realistically we might prefer to assume that (4) only holds approximately. Proceeding in this way is a central theme of Chamberlain (1994). We may explore this idea later, but for the time being assume that (4) holds.

To characterize the sampling properties of $\hat{\beta}(\tau)$ it is helpful to write (5) in matrix form. Let $\mathbf{W} = (w_1, \ldots, w_M)'$ be the $M \times K$ matrix consisting of the values of W at each of the support points in $\mathbb{X}$. We have, with all notation as defined above,

$$\begin{aligned} \min_{b \in \mathbb{R}^K} \sum \frac{\hat{p}_m}{\hat{\sigma}_m^2} \left(\hat{\pi}_m(\tau) - w_m' b\right)^2 &= \min_{b \in \mathbb{R}^K} \sum \hat{\Omega}_{mm}^{-1} \left(\hat{\pi}_m(\tau) - w_m' b\right)^2 \\ &= \left(\hat{\pi}(\tau) - \mathbf{W} b\right)' \hat{\Omega}^{-1} \left(\hat{\pi}(\tau) - \mathbf{W} b\right). \end{aligned} \tag{6}$$

Here $\hat{\Omega}_{mm}$ denotes the mth diagonal element of $\hat{\Omega} = \operatorname{diag}\{\hat{\sigma}_1^2/\hat{p}_1, \ldots, \hat{\sigma}_M^2/\hat{p}_M\}$. It is worth taking the time to verify that (5) and (6) are equivalent. To do so proceed backwards: evaluate (6) to show that it coincides with (5).

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Differentiating (6) with respect to the K elements of b yields the first order conditions

$$\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\hat{\beta}(\tau) - \mathbf{W}'\hat{\Omega}^{-1}\hat{\pi}(\tau) = 0.$$

Solving for $\hat{\beta}(\tau)$ gives

$$\hat{\beta}(\tau) = \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\hat{\pi}(\tau). \tag{7}$$

Equation (7) coincides with the weighted least squares (WLS) fit of $\hat{\pi}_m(\tau)$ onto w_m for $m = 1, \ldots, M$. However, unlike in a typical linear regression model, their is no "regression error" since

$$\pi_m(\tau) = w_m'\beta(\tau)$$

holds *exactly* for all for $m = 1, \ldots, M$. This implies that all of our sampling uncertainty about $\hat{\beta}(\tau)$ stems from our sampling uncertainty about $\hat{\pi}(\tau)$. In principle we might also be concerned about sampling uncertainty in $\hat{\Omega}$, but it turns out we can ignore this complication when (4) holds. Chamberlain (1994) shows that when the linear quantile regression function is misspecified, such that (4) no longer holds, sampling uncertainty in $\hat{\Omega}$ matters in a dramatic way. An (interesting) complication which we will ignore.

To derive the asymptotic sampling distribution of $\sqrt{N}\left(\hat{\beta}(\tau) - \beta(\tau)\right)$ under correct specification note that

$$\begin{aligned}\hat{\beta}(\tau) &= \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\hat{\pi}(\tau)\\ &= \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\pi(\tau) + \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\left(\hat{\pi}(\tau) - \pi(\tau)\right).\end{aligned}$$

Since, under (4), $\left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\pi(\tau) = \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\beta(\tau) = \beta(\tau)$ we further have

$$\begin{aligned}\sqrt{N}\left(\hat{\beta}(\tau) - \beta(\tau)\right) &= \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\sqrt{N}\left(\hat{\pi}(\tau) - \pi(\tau)\right)\\ &= \left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}\mathbf{W}'\hat{\Omega}^{-1}\sqrt{N}\left(\hat{\pi}(\tau) - \pi(\tau)\right),\end{aligned}$$

which using a large of large numbers (LLN), result (3), and a Slutsky theorem, yields

$$\sqrt{N}\left(\hat{\beta}(\tau) - \beta(\tau)\right) \xrightarrow{D} \mathcal{N}\left(\mathbf{0}, \left(\mathbf{W}'\Omega^{-1}\mathbf{W}\right)^{-1}\right). \tag{8}$$

We can estimate the asymptotic variance in (8) by $\left(\mathbf{W}'\hat{\Omega}^{-1}\mathbf{W}\right)^{-1}$ and use its diagonal elements to construct standard errors for the elements of $\hat{\beta}(\tau)$.

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References

Chamberlain, G. (1994). *Advances in Econometrics: Sixth World Congress*, volume 2, chapter Quantile regression, censoring, and the structure of wages, (pp. 171 – 209). Cambridge University Press: Cambridge.

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